Unconditional sphericality and relative-center reduction for $R6_{9,3748}$

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Outcome and scope. Every characteristic-zero fusion categorification of the specified ring admits a spherical structure. Its rank-three subcategory is one of the $\mathrm{PSU}(2)_5$ categories or Galois conjugates; the complementary bimodule is categorically regular. Consequently the previously conditional fifteen-case relative-center calculation applies without extra hypotheses. It leaves two one-sided cases, each of which gives a central lift of the rank-three subcategory. The companion `center_exclusion.tex` uses this central lift to exclude every categorification. The present note supplies its categorical reductions. No tensor-factorization theorem for an unbraided ambient category is asserted.

Input. Let A have basis $1, a, b$ with

$$a^2 = 1 + b, \qquad ab = a + b, \qquad b^2 = 1 + a + b.$$

The source ring, including all its structure constants, is

$$R = A[X]/(X^2 - 1 - QX), \qquad Q = 2 + 2a + b.$$

The source labels are $(1, a, b, X, aX, bX) = (0, 1, 5, 2, 3, 4)$; all simples are self-dual. The verifier compares the reconstructed tensor with the exact source tensor. If $\mathcal{C}$ is a categorification, let $\mathcal{D}$ be its subcategory on $1, a, b$, and let $\mathcal{M}$ be the complementary full semisimple subcategory. It is a $\mathcal{D}$-bimodule category.

Lemma 1. *Every categorification of A has a spherical structure.*

Proof. Use the sphericalization $\widetilde{\mathcal{D}}$ and signed quotient of [1, Section 5.1]. Its distinguished central involution ε is the negative lift of the unit; the signed quotient imposes $\varepsilon = -1$. Both lifts of every self-dual simple are self-dual. Choose lifts of a, b. Frobenius reciprocity and the coefficient bounds in [1, Section 5.1(b)] give the signed multiplication

$$a^2 = 1 + ub, \qquad ab = ua + vb, \qquad b^2 = 1 + va + wb, \qquad u, v, w \in \{1, -1\}.$$

Associativity of $(a^2)b = a(ab)$ forces $uw = 1$, hence $w = u$. Replacing a by va and b by ub makes all three signs positive. Thus the chosen three lifts form a fusion subcategory mapping equivalently to $\mathcal{D}$, which is precisely a spherical section by [1, Section 5.1(c)]. No pivotality assumption was made. Alternatively [1, Theorem 5.8] applies directly. □

By [1, Theorem 1.1], the category $\mathcal{D}$ is the even sl_2 category at a seventh root of unity, up to the Galois choices. It admits a nondegenerate braiding, so

$$Z(\mathcal{D}) \simeq \mathcal{D} \boxtimes \mathcal{D}^{\mathrm{rev}}, \qquad K_0(Z(\mathcal{D})) = A \otimes A.$$

Here the forgetful map is $u \otimes v \mapsto uv$. This uses a braiding on $\mathcal{D}$, not a braiding on $\mathcal{C}$.

Proposition 1. *Every categorification of R has a spherical structure.*

Proof. Consider the sphericalization of $\mathcal{C}$. The preceding lemma permits positive, fusion-closed lifts of $1, a, b$ inside it. Choose a lift X_+ and define the lifts of aX, bX by a_+X_+, b_+X_+. These are simple, since their forgetful images are simple. Associativity and the positive fusion rules of the lifted $\mathcal{D}$ then make its entire left action on these three lifts positive. All twelve simples in the sphericalization are self-dual, so its Grothendieck ring is commutative and the right action is the same.

In the signed quotient the relation is consequently

$$X^2 = 1 + (p + qa + rb)X, \qquad p, q \in \{-2, 0, 2\}, \quad r \in \{-1, 1\}.$$

Replacing the three complementary lifts simultaneously by their negative lifts replaces (p, q, r) by its negative. Thus $r = 1$ may be assumed, giving exactly nine possibilities. All remaining signed products are forced by the displayed relation and the regular action of A; no independent signed structure constants remain.

Write $S_{p,q}$ for this signed ring and R for the original ring. The sphericalization tensor is forced to be

$$\widetilde{N}^{(k,g)}_{(i,e),(j,f)} = \frac{N^k_{ij} + (-1)^{e+f+g}(S_{p,q})^k_{ij}}{2}, \qquad e, f, g \in \{0, 1\}.$$

The verifier constructs every rank-twelve tensor, checks all unit, duality, reciprocity and associativity identities, computes its exact Casimir polynomial, and factors it over $\mathbb{Q}$.

If $P_{p,q}(z)$ is the signed rank-six Casimir polynomial, the rank-twelve Casimir polynomial is

$$2^{12} P_{2,2}(z/2) P_{p,q}(z/2).$$

This equality is checked directly. Thus its formal codegrees include twice the six roots of $P_{p,q}$. The following table records the coefficient of z^5 and constant coefficient of the monic, irreducible sextic $P_{p,q}$. Full polynomials and exact nonzero division remainders are in the certificate.

p	q	$[z^5]P_{p,q}$	$P_{p,q}(0)$	d-number test
-2	-2	-189	3296573	fails
-2	0	-77	703493	fails
-2	2	-245	6754013	fails
0	-2	-105	1279733	fails
0	0	-105	434581	fails
0	2	-385	2854789	fails
2	-2	-133	1279733	fails
2	0	-245	3027661	fails
2	2	-637	5274997	passes

Each of the first eight rows already violates $P_{p,q}(0) \mid ([z^5]P_{p,q})^6$. Multiplication by the rational number two preserves the d-number property (equivalently, cancels from this divisibility criterion). Thus each row contradicts [2, Theorem 1.2 and Lemma 2.7(v)] for the actual sphericalization. The only possible row is $(p, q, r) = (2, 2, 1)$. In that case the six chosen lifts have precisely the positive original fusion rules, so are closed under tensor products and duality. They give a fusion subcategory mapping equivalently to $\mathcal{C}$, hence a spherical section. This proves the proposition. □

Galois consequence. The original formal-codegree polynomial is the irreducible sextic

$$P_{2,2}(z) = z^6 - 637z^5 + 22295z^4 - 298116z^3 + 1854944z^2 - 5274997z + 5274997.$$

For any categorification, the global dimension in its spherical structure is one of these formal codegrees. Some Galois conjugation sends that root to the Frobenius–Perron global dimension, giving a pseudo-unitary Galois conjugate. Consequently existence of a general, pivotal, spherical, or pseudo-unitary categorification of this particular based ring is equivalent. This does not say that every categorification is itself pseudo-unitary or unitary. The standard algebraic-form/Galois-conjugation framework is [5]; the dimension argument is also the one used in [1, Lemma 3.4].

Lemma 2. *The bimodule $\mathcal{M}$ is equivalent to the regular $\mathcal{D}$-bimodule, and an equivalence can be chosen to send* 1 *to* X.

Proof. The actual functor $L : \mathcal{D} \to \mathcal{M}$, $d \mapsto d \otimes X$, is a left module functor by the ambient associativity constraints. It maps the three distinct simples bijectively to the three distinct simples X, aX, bX. On simple endomorphism spaces it is the nonzero scalar map, and all other simple Hom spaces vanish. It is therefore a left-module equivalence. Equivalently its internal endomorphism algebra at X is 1.

Transporting the right action through L gives a tensor functor $\phi : \mathcal{D} \to \mathcal{D}$ (with the usual opposite-category convention for right actions), characterized by $X \otimes d \simeq L(\phi(d))$. The fusion rules make ϕ send each simple to itself. Hence ϕ is a tensor autoequivalence and $\mathcal{M}$ is the invertible twisted regular bimodule ${}_{\mathcal{D}}\mathcal{D}_{\phi}$. This proves invertibility categorically, rather than inferring it from the fusion module alone.

By [3, Theorem 1.2], $\mathrm{BrPic}(\mathrm{Ad}(A_6)) = 1$, including all its Galois conjugates. Thus $\mathcal{M}$ is equivalent to the regular bimodule. Such an equivalence sends the unit to X: if it sent it to aX or bX, left tensoring by a would send a simple to a nonsimple object, impossible for an equivalence from the regular module. □

Proposition 2. *The three categorical hypotheses in the previous conditional relative-center note hold for every categorification of* R.

Proof. Centers of bimodules are bimodule-functor categories [4, Definition 2.1]. A regular bimodule equivalence $E : \mathcal{D} \to \mathcal{M}$ induces $Z(\mathcal{D}) \simeq Z_{\mathcal{D}}(\mathcal{M})$. Its unit image is a relative-central object $t = (X, \gamma)$. With $B = A \otimes A$, the relative-center simple basis is therefore $B \sqcup Bt$, and its forgetful map sends t to X and $u \otimes v$ to uv.

For completeness, B commutes with t in the Grothendieck ring for a categorical reason. Use strict notation and the convention $\gamma_v : vt \to tv$. If $z = (d, \sigma) \in Z(\mathcal{D})$, the map $\gamma_d : dt \to td$ is a morphism in the relative center. The required identity is

$$(\gamma_d \otimes 1_v)(1_d \otimes \gamma_v)(\sigma_v \otimes 1_t) = (1_t \otimes \sigma_v)(\gamma_v \otimes 1_d)(1_v \otimes \gamma_d).$$

It follows by expanding γ_{dv} and γ_{vd} and applying naturality of γ to $\sigma_v : vd \to dv$. Thus $zt \simeq tz$. This is also the standard central action of $Z(\mathcal{D})$ on the relative center. The basis element t is self-dual: under the equivalence with $Z(\mathcal{D})$ it is the unique simple forgetting to X, since the unit is the unique simple of $Z(\mathcal{D})$ forgetting to 1.

The nonnegative B-part of t^2 forgets to 1, so it is exactly 1. The remaining coefficient is $C \in B_{\geq 0}$ with $F(C) = Q$. Hence

$$K_0(Z_{\mathcal{D}}(\mathcal{C})) = B[t]/(t^2 - 1 - Ct), \qquad F(C) = 2 + 2a + b.$$

This is precisely the finite domain of the existing fifteen-case certificate. □

Consequence of the previous arithmetic. The already verified d-number calculation eliminates thirteen of the fifteen possibilities. The only remaining possibilities are

$$C = Q \otimes 1 \quad \text{or} \quad C = 1 \otimes Q.$$

The original arithmetic is unchanged; its applicability is now unconditional. In either case the corresponding six-simple fusion subcategory of the relative center maps equivalently to $\mathcal{C}$. A tensor section $\mathcal{C} \to Z_{\mathcal{D}}(\mathcal{C})$ supplies a coherent central lift $\mathcal{D} \to Z(\mathcal{C})$: for fixed $d \in \mathcal{D}$, take its crossing with every $Y \in \mathcal{C}$ from the relative half-braiding of the section of Y. Naturality in Y and multiplicativity in Y follow from functoriality and monoidality of the section. Its restriction to $\mathcal{D}$ is one of the two nondegenerate braidings.

Role in the complete exclusion. The surviving central lift does not supply a braiding on $\mathcal{C}$. Instead the companion `center_exclusion.tex` applies the Müger factorization theorem inside the braided category $Z(\mathcal{C})$. Its modular centralizer has a dimension multiplicity bound of four, while a required thirteen-primary twist has a Galois orbit of at least six objects of the same dimension. This excludes both one-sided relative-center survivors and, by the spherical/Galois reduction above, every characteristic-zero categorification. No nongraded extension classification or associator construction is assumed.

Reproduction. Run `python verify_sphericality.py --verify`; it replays all nine rank-twelve tensors and exact polynomial/divisibility calculations. The script also succeeds with `python -O`; it has no assertion-based correctness gates. A generation-and-replay run completed in under two seconds. The files are confined to this research directory, and the original package is unchanged.

References

[1] V. Ostrik, *Pivotal fusion categories of rank 3*, Moscow Math. J. 15 (2015), 373–396, arXiv:1309.4822.

[2] V. Ostrik, *On formal codegrees of fusion categories*, Math. Res. Lett. 16 (2009), 895–901, arXiv:0810.3242.

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[4] S. Gelaki, D. Naidu, D. Nikshych, *Centers of graded fusion categories*, arXiv:0905.3117, Section 2.2.

[5] P. Etingof, D. Nikshych, V. Ostrik, *On fusion categories*, arXiv:math/0203060.